Welcome to Management Science!

Management Science

Course Description

Management Science is an interdisciplinary field that applies scientific methods to organizational problem-solving and decision-making. By leveraging mathematical modeling, statistics, and numerical algorithms, management science helps businesses achieve their strategic goals effectively.

In this course, you'll build a comprehensive toolkit by solving real problems across diverse business domains. Each algorithm is a tool, each case is a client, and each presentation is a pitch. Throughout the semester, you'll work with realistic, business-relevant scenarios using Python. The course end in a consulting competition where teams tackle client briefs (food delivery routing, healthcare staff scheduling, or inventory optimization) and present solutions to a panel of "executives."

Learning Outcomes

By the end of this course, you will be able to:

Technical Skills:

- Implement Management Science solutions in Python
- Work effectively with NumPy and Pandas for analysis and modeling
- Apply Monte Carlo simulation to model uncertainty and risk
- Build and evaluate forecasting models for demand and time series
- Design and analyze scheduling solutions with key performance metrics
- Solve routing problems using heuristics and local search improvements
- Handle multi-objective trade-offs and combine decision criteria meaningfully
- Understand and apply metaheuristics for complex optimization problems

Professional Skills:

- Collaborate effectively in small teams (3-4 students)
- Communicate technical insights with clear visualizations and compelling narratives
- Present solutions in a consulting-style format to business stakeholders
- Approach complex problems with structured analytical thinking

> **i** Note
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> This course is specifically designed for business students. No prior programming experience required, the teaching format supports different skill levels so every student can progress effectively.

Course Structure

The course is organized into three distinct parts across 12 lectures:

- **Part I:** Python Foundation (Lectures 1-3)
- **Part II:** Management Science Tools (Lectures 4-9)
- **Part III:** Consulting Competition (Lectures 10-12)

Grading

- **Assignment 1: Risk & Forecasting** (Due Lecture 8) - 30%
- **Assignment 2: Full Optimization Toolkit** (Due Lecture 10) - 30%
- **Final Competition: Client Project** (Lectures 10-12) - 40%

> Tip
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> - Win mini-competitions during lectures (Lectures 4-9)
> - Best client project in Part III chosen by peer teams

Resources & Support

Required Tools

- Laptop capable of running Python and Jupyter notebooks
- Python (installed via uv package manager in class)
- An AI coding assistant (see the AI policy below; a free Mistral account or Zed's education plan both work — GitHub Copilot student signups are currently paused)
- VS Code or similar IDE

Getting Help

1. **During class:** Ask questions immediately, others likely have the same question
2. **Team support:** Leverage your group for collaborative problem-solving
3. **Email:** Response within 48 hours for urgent questions

AI Policy

Course chatbot. A course chatbot is available on this website for exploratory study. It is designed to guide your problem-solving process rather than hand out answers: expect hints and questions back, not finished solutions. It runs on EU-hosted infrastructure (Mistral models, servers in Germany), and your conversations are not used to train models. Details are on the privacy page.

External AI tools. You may use external AI tools (e.g., ChatGPT, Claude, Mistral, Gemini). You are responsible for every line of code and every sentence you hand in: understand what the AI produced before you rely on it. Letting AI solve your tasks for you weakens exactly the skills this course exists to build, and running code you do not understand can introduce errors and security risks.

Disclosure. Every submission that used AI says so, in one honest line: what the tool was used for, in your own words (for example: *"Used AI to explain the error message and to draft the plotting code, reworked by me."*). This is not about catching anyone out; being able to say clearly what a tool did for you is part of using it well.

Exam. In the examination itself, AI tools are not permitted unless your examination regulations or the exam instructions explicitly say otherwise.

How to Navigate This Course

- **Slides:** Click "RevealJS" in the top right corner of each lecture page
- **Notebooks:** Interactive exercises accompany each lecture

You can find detailed information about course policies, grading rubrics, and expectations in the syllabus.

Questions & Contact

If you have any questions regarding the course, please contact me at vlcek@beyondsimulations.com.

Contributors

Thanks to Asvin Goel, who inspired part of this course.

Bibliography